ACE Cyber Project Research Design

1. Project Information

Field	Response
Student Name(s)	Teaundra Thompson
University	Charleston Southern University
Faculty Advisor	Professor Henderson
DoD/ACE Mentor	Professor Littleton
Project Title	IDS Resiliency Against Evasion Attacks
Version	v1.4
Date	12/17/2025

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2. Research Method Selection

Select one:

☐ Experimental ☐ Applied ☐ Descriptive ☐ Exploratory

☐ Mixed Methods

Briefly describe why this method was chosen:
(Explain how this method best aligns with your research question and objectives.)

This method was chosen because I am looking to evaluate the performance of a IDS detecting adversarial evasion attacks. The variables will be carefully controlled to thoroughly test the system.

3. Research Design Overview

Provide a high-level description of your research design:

- What are you studying?
- What are the key variables or factors?
- What is your overall approach (quantitative, qualitative, or mixed)?
- What do you expect to learn or demonstrate?

Example: "This study will use a controlled experiment to test how a conversational AI model improves log query efficiency compared to baseline manual methods."

This study will utilize a VM environment to test a baseline IDS's resiliency against evasion attacks such as payload encoding and mimicry. My overall approach will be quantitative. The baseline IDS will be created using Zeek and Google Colab will be used for creating and training a MLP ML model. The MLP model will be trained on the UFWZeekData24 dataset. Once the ML model is trained, I will feed the baseline IDS with pcap files that display different types of traffic, which will generate a connection log that the MLP model will analyze. Some of the key variables are the UFWZeekData24 dataset, the MLP model, Zeek's conn.logs, and the content of the various pcaps. This study will be quantitative, and I expect to gain data showing how vulnerable a ML-based IDS is against evasion attacks. I will also present solutions for mitigating the vulnerabilities and suggestions for increasing IDS robustness.

4. Research Objectives & Hypothesis

Primary Objective(s):

1. Quantify accuracy deterioration under evasion attacks
2. Quantify the success rate of adversary attacks
3. Define solutions for IDS hardening against evasion attacks

Hypothesis (if applicable):

If [independent variable/intervention] is introduced, then [dependent variable/outcome] will [expected result].

A five-layer Multilayer Perceptron IDS model will have a higher true positive rate against adversarial examples that have a higher frequency (large volume attacks), than examples that have a low frequency (small volume attacks).

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5. Planned Research Steps / Experimentation Plan

Outline your step-by-step approach to conducting your research.
Each step should be clear and actionable.

Step	Description	Expected Output	Duration / Timing
1	Define and refine research question	Finalized research focus	Week 1- 3
2	Conduct literature review	Literature map	Week 4-5
3	Research open-source IDS software, datasets, ML models	Research Design	Week 6-7
4	Develop prototype / model / framework	Initial build	Week 8-10
5	Ensure Zeek and MLP model work	Successful testbed setup	Week 11
5	Baseline and adversarial testing	Data Collected	Week 12
6	Analyze results	Findings Summary	Week 13
7	Prepare report & presentation	Final submission	Week 14 - 16

Adjust as appropriate for your research timeline.

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6. Data Collection Plan

Describe how you will collect your data, what type of data you expect (quantitative, qualitative, or both), and how you will ensure accuracy and reliability.

Data Source	Type (Quant / Qual)	Description	Collection Method	Tool / Instrument	Notes
Network Traffic	both	Network traffic contains benign and adversarial evasion techniques.	Generated traffic will be captured into a pcap	Wireshark	Traffic is generated by web browsing, and executing commands and/or running python scripts.
Zeek	both	Logs created from pcap files	Output from Zeek	Zeek	test with normal pcap and adversarial pcap file
MLP Model	Quantitative	MLP model ouput based on Zeek logs	Output from MLP, sent back to Zeek via API	Google Colab	Probability of-attack will-be numerically based

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7. Data Analysis Plan

Explain how you will analyze your data to test your hypothesis or address your research question.

- What statistical tests, visualizations, or coding methods will you use?

- How will you interpret qualitative responses (e.g., coding themes, content analysis)?
- How will success be measured (KPIs, accuracy, latency, efficiency, satisfaction, etc.)?

The data will be analyzed quantitatively using the accuracy rate, and the F1-score metric to evaluate the MLP model's detection performance on the Zeek logs. Zeek's performance is not being measured, only the MLP performance is being measured. Communication delays between Zeek and the MLP model will be analyzed thematically to explain why they occur. The TPR will determine the success of the test cases; a higher TPR indicates a higher success rate.

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8. Research Tools, Platforms, and Resources

List all tools, environments, or resources you will use during the project.

Tool / Resource	Purpose	Access Needed?	Notes
VirtualBox (v 7.1.6)	To Host Zeek	No	Determine physical hardware requirements
Wireshark (v 4.2.2)	To capture network traffic	No	Captured traffic will be stored in pcapng format
Google Colab	Create and train MLP model	No	The MLP will be created, trained, tested, and hosted here
Zeek (v8.0.1)	Baseline IDS	No	Can be installed in VM

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9. Risks & Mitigations

Identify potential risks or challenges that may affect the success or validity of your study, and describe how you plan to mitigate them.

Risk	Potential Impact	Mitigation Strategy
Improper-MLP integration with Zeek	May result in MLP not properly interpreting Zeek logs	Spend more time making prototype
Inability to transfer MLP from cloud to VM	Retraining MLP on local machine or VM	Research on how ML models can be transferred out of a cloud environment
MLP model overfitting on dataset	Will result in unsavory testing results.	Use no more than 20 training epochs and incorporate dropout layers into the model.
Too expensive to train MLP in cloud environment	Training MLP on local machine or VM	Reduce dataset size to 40% to mitigate costs

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10. Ethical & Data Handling Considerations

- Will your study involve human participants (e.g., surveys, interviews)?
- Will you handle sensitive or anonymized data?
- How will you ensure compliance with ethical guidelines, FERPA, or DoD data protection rules?

Ensure no personally identifiable data is used.

My study will not involve human participants, and I will be using anonymized open-source data. The dataset I will use is available publicly and contains no PII, and the pcaps I intend to use will not contain PII either. I will not deploy the IDS on a live network to prevent capture of sensitive data. Since I will not be

interacting with sensitive data, I will not need to explicitly ensure compliance with ethical guidelines.

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11. Expected Outputs

List what deliverables or measurable outputs will result from your research process.

Deliverable	Description	Target Date
Prototype IDS	IDS baseline model with performance metrics	11/9/2025
Adversarial examples	Pcaps created using Scapy/Python scripts, hping3, and nmap.	11/15/2025
Data Analysis Report	Adversarial testing report	11/28/2025
Final Presentation	15-20 minutes	12/11/2025

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12. Mentor Feedback / Sign-Off

Reviewer	Signature / Initials	Date	Comments
Faculty Advisor			
DoD/ACE Mentor			

Submission Instructions:

Upload the completed “Research Design / Methodology Outline” to your course’s assigned workspace (Blackboard) by the specified due date.